

Yuyun Yang

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SUMMARY

- Seeking software engineering opportunities. M.S. Computer Science student at Northeastern University; former AWS SDE Intern on the Aurora Zero-ETL team. Incoming IBM Intern.
- Built an AI-agent-assisted testing framework at AWS reducing test feedback time by 96%; hands-on experience in Spring Boot, distributed systems, and RAG/LLM applications.

EDUCATION

Northeastern University

Master of Science (M.S.) in Computer Science

San Jose, CA

Sep 2025 – Dec 2027

Courses: Object Oriented Programming, Natural Language Processing, Data Structures & Algorithms, Computer Systems, Discrete Structures

University of Leeds

Master of Arts (M.A.) in Marketing & Advertising

Leeds, United Kingdom

Sep 2019 – Nov 2020

Hangzhou Dianzi University

Bachelor of Engineering (B.Eng.) in Environmental Engineering

Hangzhou, China

Sep 2014 – Jun 2018

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, SQL, C

Frameworks & Libraries: Spring Boot, Spring Data JPA, Hibernate, Guice, REST APIs, JUnit, Swagger, FastAPI, Node.js, Express.js, HTML/CSS

Databases & Data: MySQL, PostgreSQL, DynamoDB, Redis, Change Data Capture (CDC), Parquet, Power BI

Cloud & Infrastructure: AWS (S3, EC2, Aurora, KMS, IAM), Google Cloud Platform (GCP), Docker, Docker Compose, Terraform, LocalStack, MiniStack, Git

AI & Machine Learning: Large Language Models (LLMs), AI Agents, RAG, NLP, PyTorch, Hugging Face Transformers

WORK EXPERIENCE

Amazon Web Services, Inc.

Seattle, WA

SDE Intern — Aurora Storage Streaming / Zero-ETL (Database)

May 2026 – Jul 2026

- Designed and built an **AI-agent-assisted local end-to-end testing framework** adopted for daily use by the Aurora Zero-ETL team, reducing test feedback time by **96%** (from ~2 hours in integration testing to **5 minutes locally**); automated environment provisioning, distributed workflow execution, and immediate validation of local code changes.
- Developed an **AI agent** equipped with local testing tools, curated context, and SOPs, with subagent delegation, to reproduce customer tickets, generate custom SQL workloads, execute end-to-end tests, root-cause failures, and implement code fixes.
- Orchestrated a **distributed data pipeline** across Java workers, C++ streaming daemon, MySQL/PostgreSQL, DynamoDB, and AWS; ran the production state machine locally with **Parallel Export** and **CDC** implementations via dependency-injection overrides.
- Implemented end-to-end happy-path validation across 4 dimensions: channel-state lifecycle transitions, Parquet output and row counts, CDC checkpoints, and injected INSERT/UPDATE/DELETE JSON event counts.

AstraZeneca

Shanghai, China

Associate Product Manager — Respiratory (Asthma) Portfolio

Mar 2021 – Aug 2024

- Developed automated Power BI dashboards integrated with MySQL queries to efficiently track market KPIs.
- Partnered with hospitals, physicians and sales teams to identify stakeholder needs, break down business requirements, map patient journeys, and build a data-driven care management system.
- Collaborated with the eCommerce team to design UI/UX of product pages, increasing page views by **30%**.

PROJECTS

F-1 Immigration Legal RAG System

Feb 2026 – Apr 2026

- Engineered a **Retrieval-Augmented Generation system** over a **10K-chunk** F-1 legal corpus, enabling source-grounded answers from official university and government documents.
- Improved retrieval performance from **50.7% to 84.0% Hit@5** — a **66% relative increase** — by optimizing document chunking, metadata filtering, and vector indexing on a 75-question evaluation set with LLM-assisted, human-verified ground truth.
- Deployed the Dockerized application on Google Cloud Platform using **Terraform-based Infrastructure as Code**, enabling reproducible cloud provisioning and deployment.

Patient Triage Appointment Platform

Oct 2025 – Nov 2025

- Developed a patient triage and scheduling platform for dental clinics, streamlining symptom intake, patient data management, and scheduling workflows.
- Designed a **layered Spring Boot backend architecture** with secure REST APIs, supporting role-based access control for patient onboarding, appointment creation/update/cancellation, and time-conflict validation.
- Containerized the application and PostgreSQL using Docker Compose, implemented persistence with Spring Data JPA/Hibernate, and added JUnit tests and Swagger to improve reliability and maintainability.